

Selective Passivation Evidence

Thirty-three simulations testing one question: can a robot doing sustained contact work still carry a hard, checkable bound on the energy it delivers to a human?

Plant MuJoCo 3.11.0, 1 kHz Controllers 5 Scenarios 6 + $\hat{\mu}$ sweep Runs 33
Criteria 43/43 pass

THE CLAIM UNDER TEST

Exempting predicted task power from the energy accounts is what makes a human-port guarantee compatible with contact work.

A conventional passivity tank monitors the whole measured port, $f_{\text{meas}}^\top v$. During ordinary frictional sliding that quantity is persistently negative, so the tank drains on task work the robot is *supposed* to do. The proposal instead subtracts a nominal contact-model prediction and monitors only the residual $f_R = f_{\text{meas}} - \hat{f}_T$, with a second, smaller account tracking the human channel alone.

The POC's job is to establish that this separation actually buys what it promises — and to find where it stops working. Both happened.

WHAT WAS SIMULATED

PLANT	Two-link planar arm (0.6 m links), spherical pad on a horizontal surface, MuJoCo contact as ground truth. Only the pad–surface pair can collide.
TASK	Regulate $F_N = 5 \text{ N}$ while sliding at $v_t = 0.05 \text{ m/s}$. True friction $\mu = 0.30$, 1 ms control period.
CHANNELS	Task contact f_T from MuJoCo; human force f_H applied at the end-effector site as a scripted trapezoid. Both retained separately as ground truth — the POC assumes perfect separation by design.

ACCOUNTS	Residual E_R (0.30, 0.02, 0.50) J and human E_H (0.05, 0.005, 0.08) J, giving robot-output budgets $B_R = 0.28$ J and $B_H = 0.045$ J.
CONTROLLERS	C0 nominal (no passivation) · C1 whole-port scalar · C2 residual-only QP · C3 dual-ledger QP (proposed) · C4 dual-ledger scalar (task-preservation ablation).
SCENARIOS	A exact model, no human · B model mismatch, no human · C blocking human · D mismatch + blocking · E oblique human · F helping then blocking · plus a $\hat{\mu}$ sweep.

Baseline fidelity: steady sliding holds F_N at 4.995 N (0.1 % error) with contact active 100 % of steps and 1.33 mm penetration, so the controller being filtered is a working one, not a strawman.

FINDINGS

FINDING 1 – THE PRINCIPAL RESULT

The human-port guarantee is completely insensitive to contact-model quality

- Established, exactly

Sweeping the predictor's friction coefficient across $\hat{\mu} \in \{0.20, 0.25, 0.30, 0.35\}$ against a true $\mu = 0.30$, every human-port quantity is **bit-identical**, while the residual account absorbs the entire span of model error.

E_H FLOOR REACHED 0.0050964 J Identical at all four $\hat{\mu}$ · floor is 0.005 J	CUMULATIVE HUMAN EXTRACTION 0.0449036 J Identical at all four $\hat{\mu}$ · bound is 0.045 J
RESIDUAL BUDGET REMAINING 0.043 → 0.300 J Absorbs the whole span instead	RECOVERY AFTER RELEASE 0.655 s Identical at all four $\hat{\mu}$

This is the decoupling the two-account structure exists to produce. It is worth being precise about why it is not circular: the predictor only ever sees the *task* channel and the end-effector velocity, so it structurally cannot explain away human force. Model error therefore has nowhere to go except E_R .

FINDING 2 – THE MOTIVATING FAILURE, QUANTIFIED

A whole-port tank drains on ordinary

friction, with no human and a perfect model

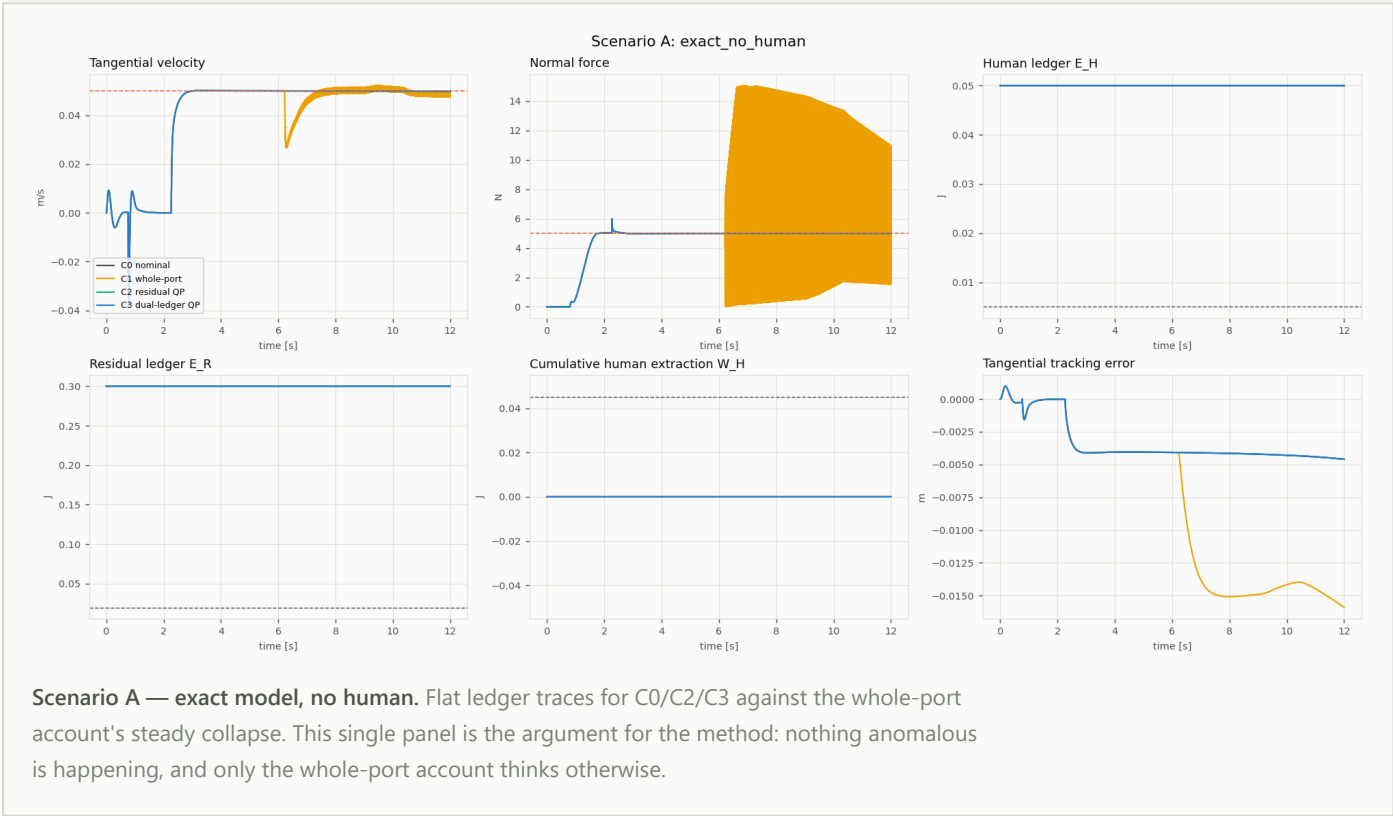
- Confirmed — and worse than the prose suggests

Whole-port ... demonstrates false depletion during ordinary frictional sliding.

Formulation §9

Measured in scenario A, where nothing anomalous happens at all: the whole-port account discharges at **0.0719 W** against the theoretical $\mu F_N v_t = 0.075 \text{ W}$, crosses its 0.02 J floor **4.0 s into sliding** ($t = 6.23 \text{ s}$, sliding begins at 2.26 s), and ends at -0.396 J — having "spent" 0.70 J that was never anyone's to spend.

Enforced rather than merely observed (C1), it produces a controller that fails on every axis simultaneously: normal-force RMSE 5.24 N against a 5 N setpoint, and 2889 infeasible steps out of 11 240. The comparison that matters for the paper is the intervention time: the residual account first constrains the command at **14.73 s** where the whole-port account does so at **6.21 s** — 8.5 s of additional useful work, bought purely by exempting predicted task power.



FINDING 3 – THE GUARANTEE ITSELF

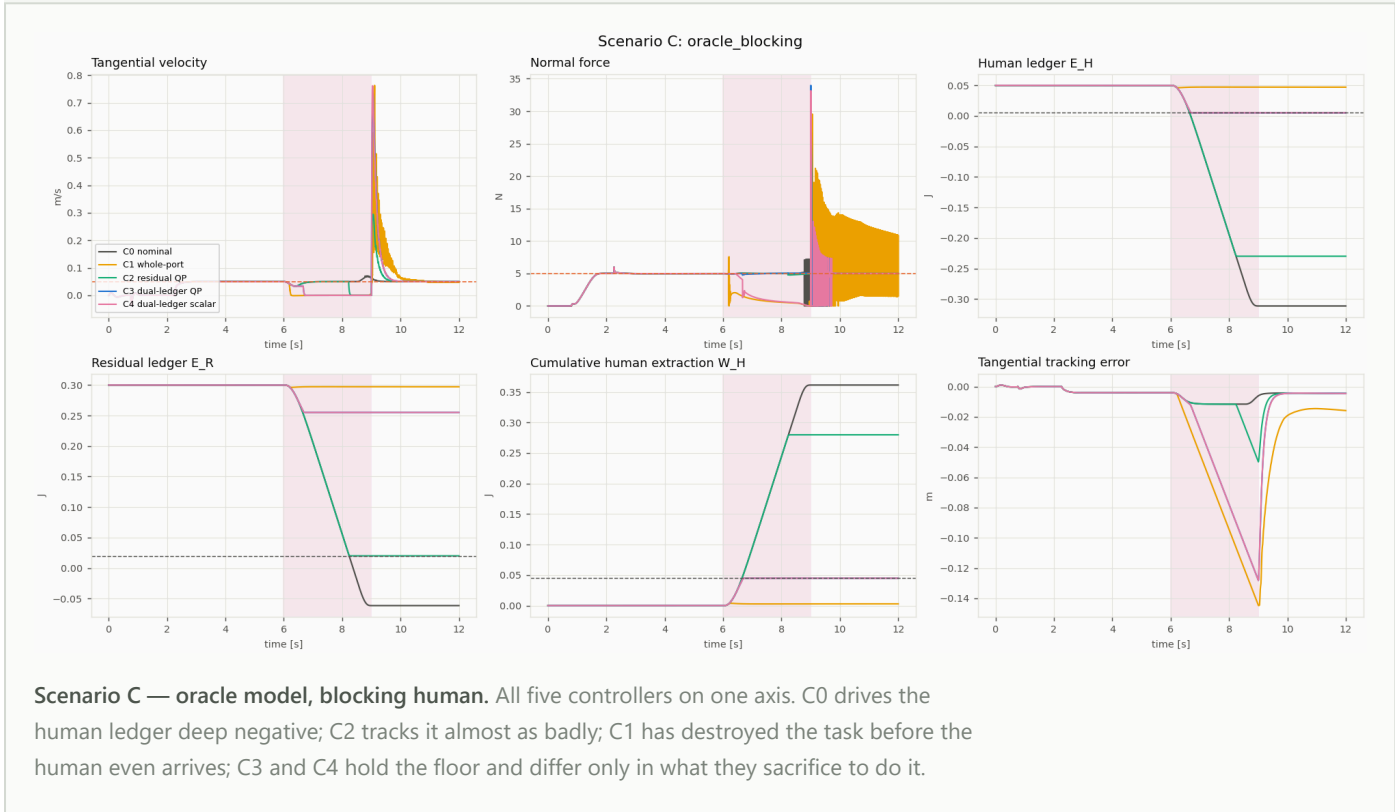
Both passivity certificates hold strictly, in every scenario

- Established

Under a sustained -3 N blocking force, the unfiltered nominal controller overdraws the human port by **0.361 J** — **$8.0\times$ the 0.045 J budget** — and would continue for as long as the human keeps pushing. The proposed controller holds both bounds with positive margin:

CERTIFICATES UNDER BLOCKING INTERACTION · EQ. (38) BOUND 0.045 J · EQ. (39) BOUND 0.28 J						
Controller	E_H min	W_H max	Eq. (38) margin	Eq. (39) margin	peak $-p_H$	Infeasible
C0 nominal	-0.3114	0.3614	-0.316	-0.081	0.1499	0
C1 whole-port	0.0462	0.0038	+0.041	+0.276	0.0380	1536
C2 residual-only	-0.2299	0.2799	-0.235	+0.0001	0.1499	0
C3 dual-ledger	0.005096	0.044904	+9.6e-05	+0.235	0.0997	0
C4 dual scalar	0.005098	0.044902	+9.8e-05	+0.235	0.0998	5

C1's apparently healthy human column is an artefact worth having ready: it never delivers energy to the human because it has already stopped doing the task. C2 is the informative row — residual-only protection leaves the human port entirely exposed ($6.2\times$ budget), exactly as §9 predicts, because $B_R = 0.28\text{ J}$ is simply too coarse to notice a 0.045 J human interaction.



FINDING 4 – WHY THE QP, NOT A SCALAR

Task-space weighting preserves the contact task $67\times$ better at identical guarantee

- Established, and the strongest argument for the QP formulation

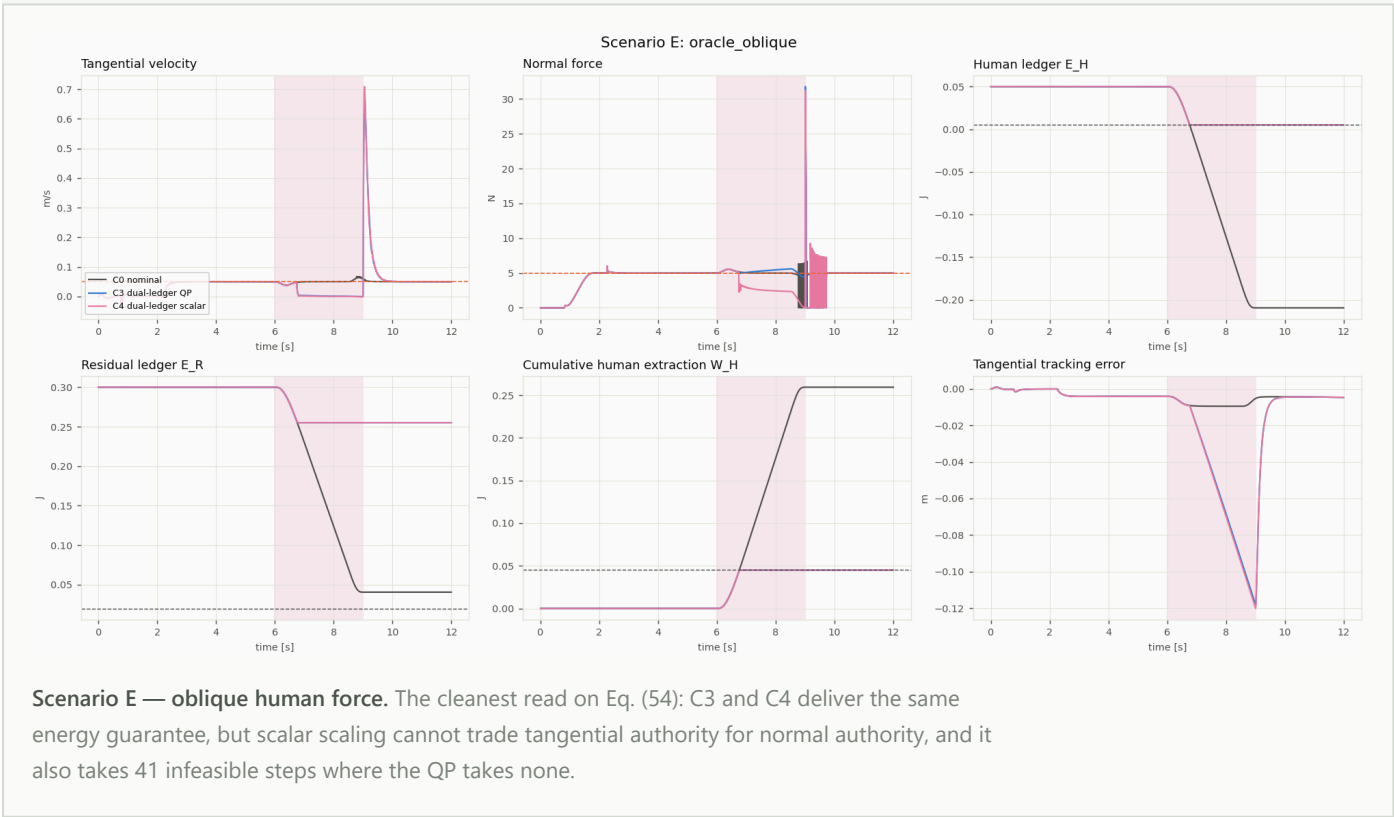
Changing the normal command is therefore twenty times as expensive as an equal change in the tangential command.

Formulation §7, Eq. (54)

C4 is the clean ablation: identical constraint rows, identical energy guarantee, but the command restricted to a scalar $u = \gamma \ u_{nom}$. The two deliver the same human-port protection to five digits. They differ entirely in what the task looks like while delivering it.

BLOCKING (C) — F_N RMSE IN WINDOW 0.055 vs 3.696 N C3 vs C4 · 67× better	OBLIQUE (E) — F_N RMSE IN WINDOW 0.378 vs 2.436 N C3 vs C4 · 6.4× better
C4 INFEASIBLE STEPS (C / E) 5 / 41 C3: zero in both — richer feasible set	C3 VS UNFILTERED C0 IN WINDOW 0.055 vs 0.349 N Better contact regulation than no filter

Two points here are worth raising deliberately. First, the oblique case is the cleanest test: with the human force split across both axes, scalar scaling has no way to concede tangential authority while retaining normal authority, so the cost lands directly on the contact force it was supposed to protect. Second — the last tile — during the interaction window C3 regulates normal force *better than the unfiltered controller*. The human force in C is tangential and perturbs F_N through arm dynamics; C3 gives up tangential tracking, which also weakens that coupling. The guarantee is not purely a tax on the task.

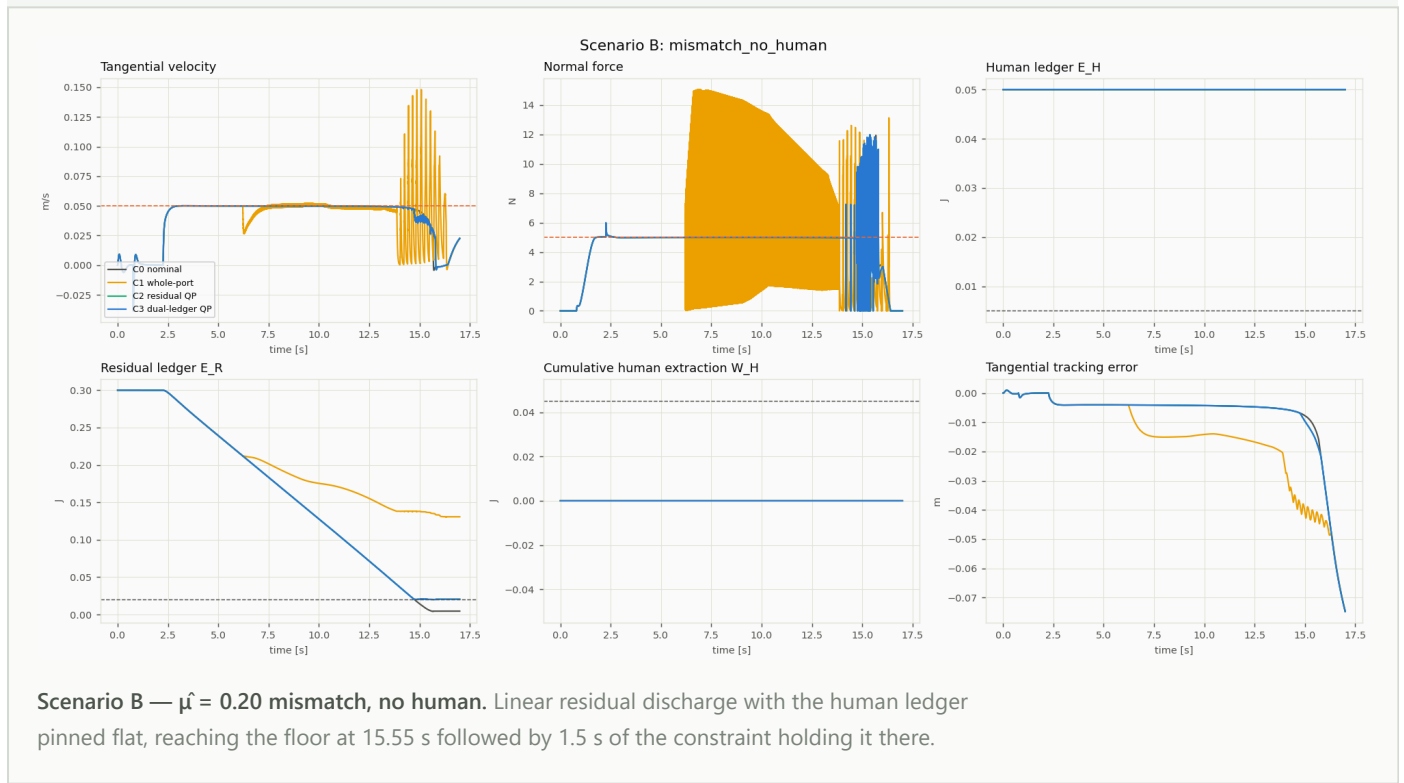


Only the residual account responds to mismatch, at the predicted rate

- Confirmed quantitatively

With $\hat{\mu} = 0.20$ against $\mu = 0.30$ and no human present, the residual account discharges linearly at **0.02238 W** against the predicted $\Delta\mu \cdot F_N \cdot v_t = 0.025 \text{ W}$ — the 10 % gap is the tanh regularisation near the sign change plus normal-force ripple. The human account stays at exactly 0.050000000000 J for the entire run. Model error has no path into the human budget.

In scenario A, where the model is exact, both accounts are constant to twelve decimal places — $E_R \in [0.300000000000, 0.300000000000]$, $E_H \in [0.050000000000, 0.050000000000]$. That is not "small drift": with an oracle predictor f_R is identically zero and the update adds exact zeros. The filter intervenes on exactly 1 step out of 11 239 and costs 0.02 % of normal-force RMSE, so the layer is genuinely transparent when it should be.



The hard certificate has a finite model-error envelope

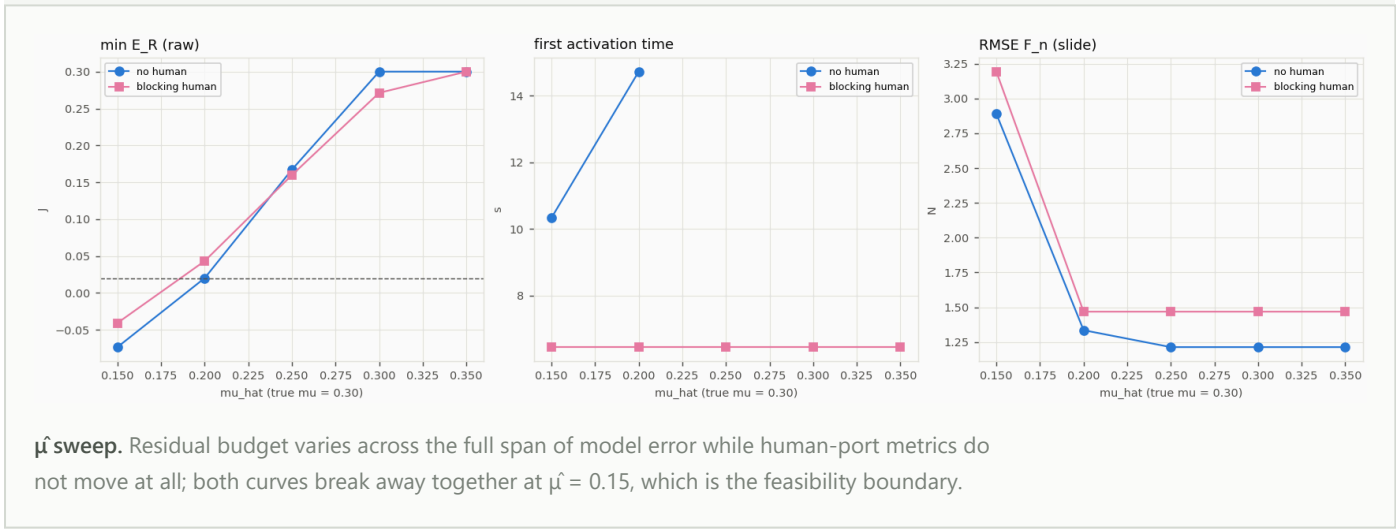
- New — not anticipated in the formulation

§7 notes that if actuator limits make the QP infeasible, an implementation must invoke a separately certified emergency action or report loss of the guarantee. The sweep gives that statement a number. At $\hat{\mu} \in \{0.20 \dots 0.35\}$ every run is feasible throughout. At $\hat{\mu} = 0.15$ the problem crosses a boundary and the guarantee does not degrade — it **fails**:

M SWEEP, CONTROLLER C3 · TRUE M = 0.30 · THE TOP ROW IS PAST THE FEASIBLE BOUNDARY							
$\hat{\mu}$	Human	Infeasible steps	E_R min	Eq. (39) margin	E_H min	W_H max	
0.15	no	1739	-0.0734	-0.0934	0.0500000	0.000	
0.15	yes	1266	-0.0407	-0.0607	0.0050964	0.044904	
0.20	yes	0	0.0432	+0.0232	0.0050964	0.044904	
0.25	yes	0	0.1598	+0.1398	0.0050964	0.044904	
0.30	yes	0	0.2712	+0.2512	0.0050964	0.044904	
0.35	yes	0	0.3000	+0.2800	0.0050964	0.044904	

The boundary sits near $|p_\Delta| \approx 0.035$ W for this plant. The mechanism is not numerical: the rate at which mismatch drains the account and the correction needed to hold the floor are both set by the same p_Δ , so they cannot be traded against each other. Past the boundary the QP is infeasible, the uncertified emergency fallback takes over, and E_R is driven below its floor.

Note the second row carefully, because it is the more interesting one: even past the feasible boundary, the *human* columns remain exactly at their nominal values. The residual certificate fails while the human certificate survives. That is suggestive of real robustness in the nesting — the small account is the one that binds first and stays bound — but it is one data point, not a proof, and would be worth establishing properly.



FINDING 7 – A GAP IN THE FORMULATION

Equation (55) as written is not realizable at 1 kHz

- Requires a change to the paper

The implementation carries CBF-style smoothing rows $-p^+ \leq k_{\text{cbf}}(E - E_{\text{min}})$ with $k_{\text{cbf}} = 50 \text{ s}^{-1}$ that do not appear in Eq. (55). I removed them to check whether they were engineering decoration. They are not:

CONTROLLER C3 WITH THE CBF ROWS DISABLED (ALL ELSE EQUAL)						
Scenario	Infeasible steps	E_H min	E_R min	Eq. (38) margin	With CBF rows	
B mismatch	788	0.050000	0.019372	+0.045	0 infeasible, floors held	

Scenario	Infeasible steps	E_H min	E_R min	Eq. (38) margin	With CBF rows
C blocking	5	0.004841	0.254841	-1.59e-04	0 infeasible, floors held
D mismatch + block	5	0.004841	0.042452	-1.59e-04	0 infeasible, floors held
E oblique	14	0.004394	0.254394	-6.06e-04	0 infeasible, floors held
F helping + block	5	0.004797	0.300000	+0.045	0 infeasible, floors held

The one-step energy rows alone collide with the torque-rate limit as the account approaches its floor: the constraint demands an instantaneous stop to extraction that the actuator cannot deliver in one 1 ms step. The CBF row makes the approach exponential with time constant $1/k_{cbf}$, so that demand never arises. Because it is strictly *tighter* than the mandated row near the floor, the §8 certificate is preserved — but it belongs in Eq. (55), not in the implementation, since its necessity cannot be seen from the equation as printed.

Two smaller items in the same category. The implemented QP also carries an instantaneous human-power row $-p_H \leq 0.10$ W that is absent from Eq. (55); it is another tightening, and it is what sets the measured peak of 0.0997 W in every human scenario (C0, without it, reaches 0.1499 W). And Eq. (39) is stated as a certificate but, unlike Eq. (38), had no verification anywhere in the pipeline — worth making symmetric in the write-up.

FINDING 8 – THE COST

The guarantee is paid for in tangential tracking and recovery time

- Real, and should be presented as a trade

TANGENTIAL RMSE IN WINDOW (C) 0.044 vs 0.0076 m/s C3 vs C0 · 5.9× worse	F_N RMSE, WHOLE SLIDE (C) 1.468 vs 0.219 N Engage/release transients
RECOVERY AFTER RELEASE (C) 0.655 vs 0.158 s 4.1× slower back to spec	COMMAND DEVIATION $\int \ U - U_{NOM}\ ^2 DT$ 2312 vs 0 The filter is working, not idling

Giving up tangential tracking is not a side effect — it is the mechanism. Against a sustained block, the only way to stop delivering energy is to stop pushing through it. The honest framing for a reviewer is that the relevant comparison is not C3 against an unguarded controller, but C3 against the alternatives that offer any guarantee at all: C1 reaches 4.05 N in the human window with the task already destroyed, C2 offers no human protection, and C4 reaches 3.70 N with 5 and 41 infeasible steps of its own.

WHAT THIS DOES NOT ESTABLISH

The scope limits below are the ones most likely to come up first, and none of them are incidental — several are load-bearing for whether the method transfers.

1. **Channel separation is assumed, not solved.** f_T and f_H are perfect ground truth by construction, and the predictor is handed the true normal task force. The hard problem — recovering the residual from one noisy measurement — is untouched. Every account here is driven by an exact residual; a real estimator makes f_R noisy and biased, and both accounts integrate that error directly. This is the single largest gap between the POC and a usable method.
2. **No sensor noise anywhere.** Related but distinct: even with perfect separation, force and velocity noise would enter the port powers quadratically in the worst case and drain accounts on noise alone.
3. **Two DOF, non-redundant, planar, single contact point.** The $W = \text{diag}(1, 20)$ argument is about trading two scalar authorities. Whether it generalises to a spatial wrench–twist port with a redundant arm is open, and Finding 4 is the result most likely to change shape there.
4. **Soft contact.** 1.33 mm steady penetration, deliberately, to keep the 1 kHz delayed force loop free of micro-bouncing. A stiffer surface would need force filtering or a faster inner loop, and filtering would put a lag directly inside the certified accounting path.
5. **The emergency fallback is reported, not certified.** §7 requires a separately certified emergency action *or* an explicit declaration that the guarantee is lost. The implementation does the second, not the first. This only matters at the boundary in Finding 6 — C3 records zero infeasible steps in all six scenarios.
6. **Timing is not a real-time claim.** QP solve is 0.108–0.118 ms median against a 1 ms period, $p_{99} \leq 0.320$ ms, but the observed maximum reaches 1.701 ms. These are Python-level wall-clock measurements on Windows including solver outliers; the tail needs proper characterisation before any hardware statement.
7. **One-step force delay.** Contact force is read with a documented 1 ms delay. The account update uses the held sample with the actual next velocity, so the discrete bookkeeping is self-consistent, but it is not continuous-time work.

QUESTIONS THE RESULTS RAISE

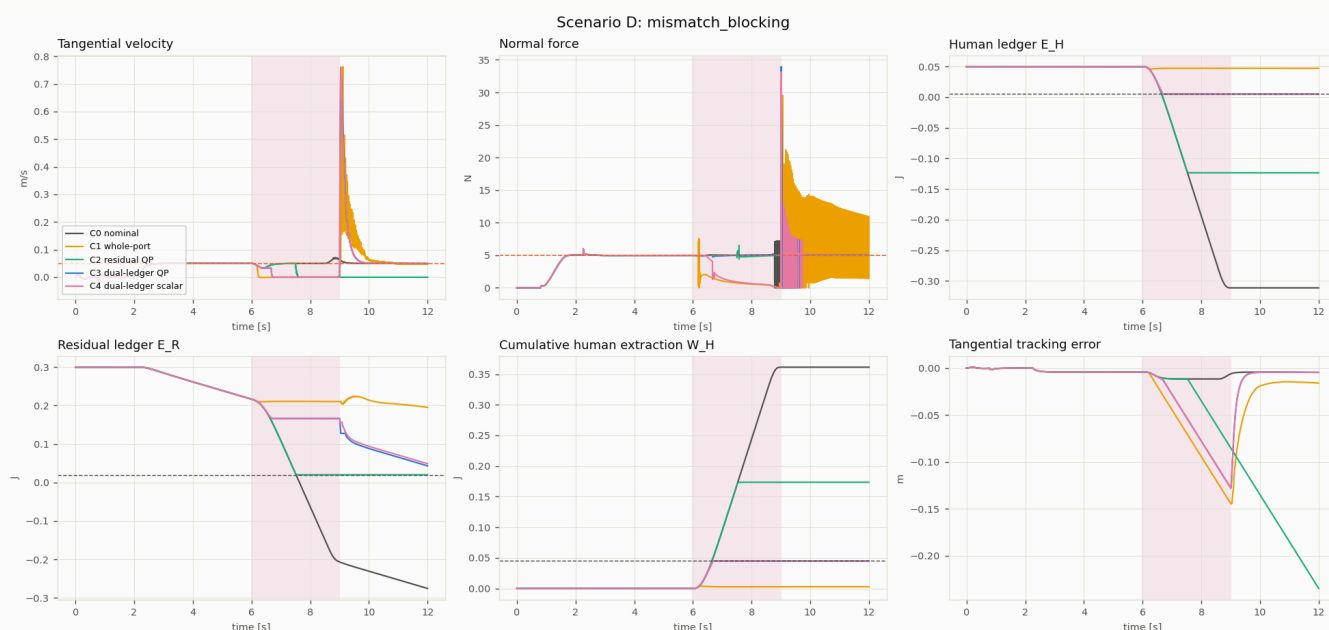
Does the nesting survive a real estimator? Finding 6's second row is the tantalising one — the human certificate held even where the residual certificate failed. If that is a structural property of nesting a small account inside a large one rather than an accident of this scenario, it is a much stronger claim than the POC currently makes, and it is exactly the property you would want when the residual is noisy.

Can the feasibility envelope be predicted rather than measured? $|p_{\Delta}| \approx 0.035$ W came out of a sweep. A bound in terms of torque-rate limit, mass matrix, and Jacobian would turn Finding 6 from a caveat into a design rule — and would tell you how to size B_R and k_{cbf} for a given arm.

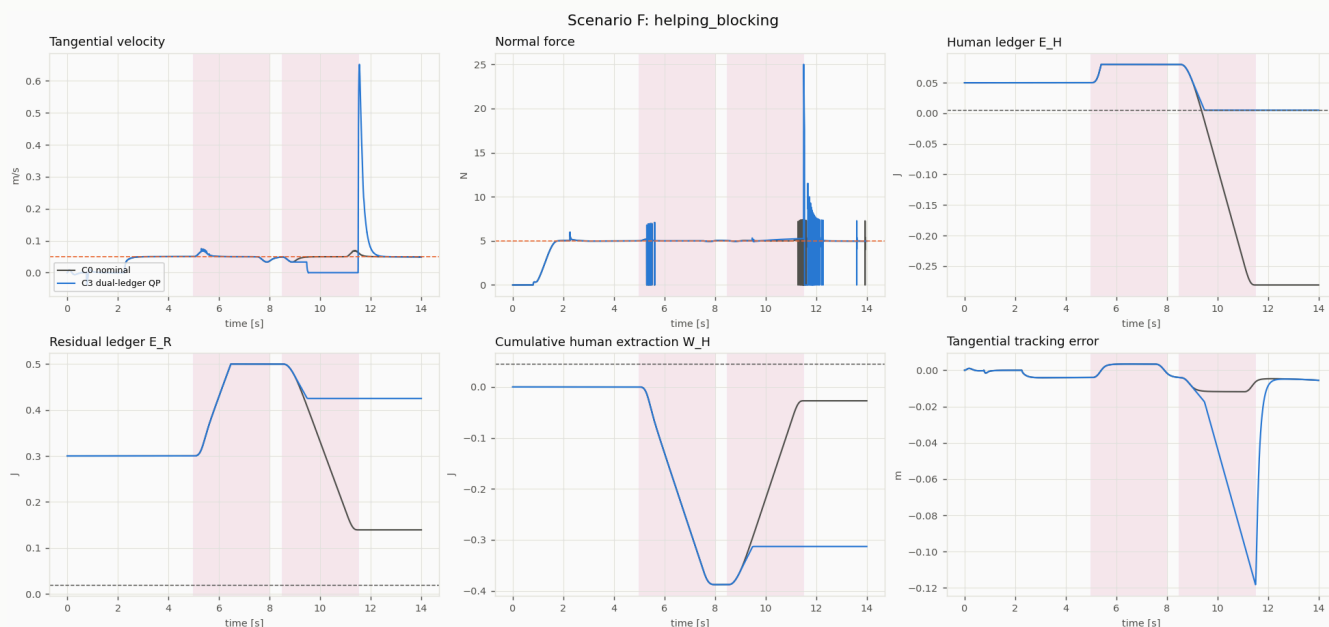
Is the whole-port comparison fair enough? C1 is the standard construction, and it fails badly. A reviewer may ask whether a better-tuned whole-port baseline exists — for instance one that also exempts a friction model but keeps a single account. That controller sits between C1 and C2 and is not currently in the comparison set.

ALL 23 SCENARIO RUNS · C3 HIGHLIGHTED · HUMAN-WINDOW COLUMNS BLANK WHERE NO HUMAN IS SCRIPTED												
Run	F _N RMSE		F _N RMSE (human)		peak -p _H	W _H max	E _H min	E _R min	E _W min	QP active	Infeas.	Recovery
A · C0 nominal	0.0348		—		0.000	0.000	0.0500	0.3000	-0.396	0 %	0	—
A · C1 whole-port	5.2372		—		0.000	0.000	0.0500	0.3000	-0.433	40.3 %	2889	—
A · C2 residual	0.0348		—		0.000	0.000	0.0500	0.3000	-0.396	0.01 %	0	—
A · C3 dual	0.0348		—		0.000	0.000	0.0500	0.3000	-0.396	0.01 %	0	—
B · C0 nominal	1.2136		—		0.000	0.000	0.0500	0.0041	-0.647	0 %	0	—
B · C1 whole-port	4.8944		—		0.000	0.000	0.0500	0.1300	-0.605	52.1 %	5725	—
B · C2 residual	1.3352		—		0.000	0.000	0.0500	0.0201	-0.633	2.4 %	0	—
B · C3 dual	1.3352		—		0.000	0.000	0.0500	0.0201	-0.633	2.4 %	0	—
C · C0 nominal	0.2193		0.3486		0.1499	0.3614	-0.3114	-0.0614	-0.754	0 %	0	0.158 s
C · C1 whole-port	3.8360		4.0457		0.0380	0.0038	0.0462	0.2962	-0.265	49.0 %	1536	none
C · C2 residual	0.6498		0.0804		0.1499	0.2799	-0.2299	0.0201	-0.655	7.0 %	0	0.563 s
C · C3 dual	1.4684		0.0548		0.0997	0.0449	0.005096	0.2551	-0.338	22.8 %	0	0.655 s
C · C4 dual scalar	2.3617		3.6962		0.0998	0.0449	0.005098	0.2551	-0.379	22.8 %	5	0.726 s
D · C0 nominal	0.2193		0.3486		0.1499	0.3614	-0.3114	-0.2756	-0.754	0 %	0	0.158 s
D · C1 whole-port	3.8360		4.0457		0.0380	0.0038	0.0462	0.1954	-0.265	49.0 %	1536	none
D · C2 residual	0.0787		0.1234		0.1498	0.1735	-0.1235	0.0201	-0.246	39.9 %	0	none
D · C3 dual	1.4684		0.0548		0.0997	0.0449	0.005096	0.0432	-0.338	22.8 %	0	0.655 s
D · C4 dual scalar	2.3617		3.6962		0.0998	0.0449	0.005098	0.0490	-0.379	22.8 %	5	0.726 s
E · C0 nominal	0.2536		0.4273		0.1067	0.2594	-0.2094	0.0406	-0.650	0 %	0	0.079 s
E · C3 dual	1.3954		0.3781		0.0997	0.0449	0.005097	0.2551	-0.329	20.4 %	0	0.724 s
E · C4 dual scalar	1.8435		2.4355		0.0997	0.0449	0.005098	0.2551	-0.335	20.4 %	41	0.713 s
F · C0 nominal	0.2859		0.3558		0.1497	0.000	-0.2809	0.1392	-0.506	0 %	0	0.180 s
F · C3 dual	1.1178		0.2532		0.0997	0.000	0.005096	0.3000	-0.136	19.3 %	0	0.774 s

Energies in J, forces in N, power in W. E_W is the conventional whole-port account, *observed* in every run and *enforced* only in C1 — that column is the false-depletion measurement. Bounds: $E_H \geq 0.005$, $E_R \geq 0.02$, $W_H \leq 0.045$, $W_R \leq 0.28$, peak $-p_H \leq 0.10$. Scenario F's W_H is zero because the helping pulse charges the account before the blocking pulse drains it, so net extraction never rises above zero.



Scenario D — mismatch + blocking, the hardest case. C3's human-port behaviour is identical to scenario C; only its residual account sits lower (0.043 vs 0.255 J). That is the division of labour the two-account structure is for.



Scenario F — helping, then blocking. Charging into the 0.08 J cap, then discharge to the floor. The flat top is forfeited surplus: the cap makes the account deliberately non-conservative, so a cooperative human cannot be banked into licence to push harder later.

33 simulations · 23 scenario runs across controllers C0-C4 · 10 μ sweep runs · ≈ 500 s simulated at 1 kHz · deterministic (fixed keyframe, scripted forces, no RNG).

Verified on MuJoCo 3.3.1 and 3.11.0, which agree to $\approx 1e-12$; 46/46 unit tests pass on both. Source data: results_passivation/summary.json, acceptance.json, sweep/sweep.csv.